

Lecture Notes on Thermodynamic Computing

Chapter 5: Thermal Noise Processing Systems

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1 5.1 Introduction to Thermal Noise Processing

• Thermal Noise as Information Carrier

Definition 1.1 (Thermal Noise Processing). Signal processing methodology that utilizes thermal noise as a computational resource rather than treating it as a detrimental effect.

Fundamental noise sources:

- Johnson-Nyquist noise: $\langle V_n^2 \rangle = 4k_B T R \Delta f$
- Shot noise: $\langle I_n^2 \rangle = 2q I_{dc} \Delta f$
- Flicker noise: $\langle V_n^2 \rangle \propto 1/f$
- Thermal-phonon interactions

• Signal-to-Noise Ratio in Thermal Regime

Traditional SNR limitation:

$$SNR = \frac{P_{signal}}{P_{noise}} = \frac{V_{signal}^2}{4k_B T R \Delta f} \quad (1)$$

Thermodynamic computing advantage:

$$\text{Information} \propto \log_2(1 + SNR) + H_{noise} \quad (2)$$

where H_{noise} is exploitable noise entropy.

• Dithering and Stochastic Resonance

Enhanced signal detection through noise:

$$\text{Output} = \text{Threshold}(V_{signal} + V_{noise}) \quad (3)$$

Optimal noise level for maximum information transfer:

$$\sigma_{opt} = \frac{V_{threshold} - V_{signal}}{\sqrt{2 \ln(2)}} \quad (4)$$

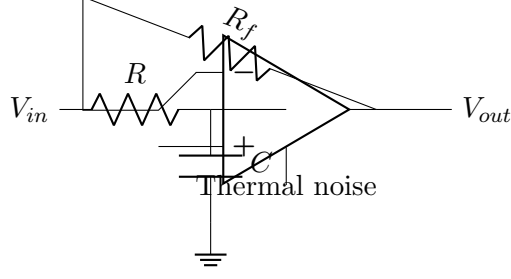


Figure 1: Stochastic resonance amplifier

2 5.2 Noise-Enhanced Signal Processing

• Stochastic Resonance Circuits

Bistable system with thermal activation:

Transfer function with noise:

$$H(\omega) = \frac{1 + j\omega RC}{1 + j\omega RC + \eta(t)} \quad (5)$$

where $\eta(t)$ is thermal noise contribution.

• Noise-Assisted Analog-to-Digital Conversion

Oversampling with dithering:

$$y[n] = \text{Quantize}(x[n] + w[n]) \quad (6)$$

Thermal noise variance:

$$\sigma_w^2 = \frac{k_B T}{C_{\text{sample}}} \quad (7)$$

Effective number of bits (ENOB):

$$\text{ENOB} = \log_2 \left(\frac{V_{FS}/2}{\sigma_{\text{quantization}}} \right) \quad (8)$$

• Subthreshold Analog Computing

Transconductance in weak inversion:

$$g_m = \frac{I_{ds}}{nV_T} \quad (9)$$

Noise-limited resolution:

$$\Delta V_{\min} = 4 \sqrt{\frac{k_B T}{C_{\text{total}}}} \quad (10)$$

Dynamic range:

$$DR = 20 \log_{10} \left(\frac{V_{\max}}{\Delta V_{\min}} \right) \quad (11)$$

3 5.3 Thermal Sampling and Quantization

- **Brownian Motion Sampling**

Random walk process:

$$x[n+1] = x[n] + \sqrt{\frac{2k_B T \Delta t}{m\gamma}} \cdot \xi[n] \quad (12)$$

where γ is damping coefficient and $\xi[n] \sim \mathcal{N}(0, 1)$.

Sampling theorem for thermal noise:

$$f_{sample} > 2 \times \frac{k_B T}{h} = 2 \times 6.25 \text{ THz at } 300\text{K} \quad (13)$$

- **Thermal Quantization**

Energy-based quantization levels:

$$E_k = k \cdot k_B T, \quad k = 0, 1, 2, \dots \quad (14)$$

Quantization error:

$$\epsilon_q = \frac{E_{input} - E_k}{k_B T} \quad (15)$$

Optimal quantizer design:

$$\Delta E = \sqrt{12} \cdot \sigma_{thermal} = \sqrt{12 k_B T} \quad (16)$$

- **Sigma-Delta Modulation with Thermal Noise**

First-order modulator:

$$y[n] = \text{sgn}\left(x[n] + \int_{-\infty}^n e[k] dk + w[n]\right) \quad (17)$$

Noise transfer function:

$$NTF(z) = 1 - z^{-1} \quad (18)$$

In-band noise power:

$$P_{noise, IB} = \frac{\sigma_w^2 \pi^2}{3} \cdot \left(\frac{f_s}{2f_{BW}}\right)^{-3} \quad (19)$$

4 5.4 Correlation and Spectral Analysis

- **Thermal Cross-Correlation**

Cross-correlation function:

$$R_{xy}(\tau) = \langle x(t)y(t+\tau) \rangle \quad (20)$$

For thermal sources:

$$R_{thermal}(\tau) = \sigma^2 \delta(\tau) + R_{signal}(\tau) \quad (21)$$

Hardware correlator using analog multipliers:

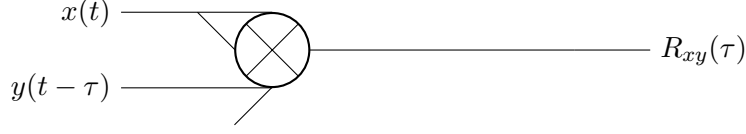


Figure 2: Thermal cross-correlator

• Power Spectral Density Estimation

Welch's method with thermal noise:

$$\hat{S}_{xx}(f) = \frac{1}{K} \sum_{k=0}^{K-1} |\text{FFT}(x_k(t))|^2 \quad (22)$$

Thermal noise floor:

$$S_{thermal}(f) = 4k_B T R \quad (23)$$

Dynamic range limitation:

$$DR_{max} = 10 \log_{10} \left(\frac{P_{signal,max}}{4k_B T R \Delta f} \right) \quad (24)$$

• Coherence Function Analysis

Magnitude-squared coherence:

$$\gamma_{xy}^2(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)} \quad (25)$$

Thermal decorrelation effects:

$$\gamma_{thermal}^2 = \frac{S_{signal}^2}{(S_{signal} + S_{thermal})^2} \quad (26)$$

5 5.5 Filtering in Thermal Environments

• Optimal Filtering with Thermal Noise

Wiener filter design:

$$H_{opt}(\omega) = \frac{S_{xs}(\omega)}{S_{xx}(\omega)} \quad (27)$$

where S_{xs} is cross-spectral density and S_{xx} is input power spectral density.
Including thermal noise:

$$S_{xx}(\omega) = S_{signal}(\omega) + S_{thermal}(\omega) \quad (28)$$

• Kalman Filtering with Thermal Uncertainty

State equation:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{w}_k \quad (29)$$

Process noise covariance:

$$\mathbf{Q} = E[\mathbf{w}_k \mathbf{w}_k^T] = \frac{k_B T}{\Delta t} \mathbf{I} \quad (30)$$

Thermal observation noise:

$$\mathbf{R} = \frac{4k_B T R}{C_{sensor}} \mathbf{I} \quad (31)$$

- **Adaptive Filtering**

LMS algorithm with thermal constraints:

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \mu e_k \mathbf{x}_k \quad (32)$$

Thermal noise limits convergence:

$$\mu_{max} = \frac{2}{\text{tr}(\mathbf{R}_{xx}) + \sigma_{thermal}^2} \quad (33)$$

Steady-state mean square error:

$$\text{MSE}_\infty = \sigma_{thermal}^2 \left(1 + \frac{\mu \text{tr}(\mathbf{R}_{xx})}{2} \right) \quad (34)$$

6 5.6 Nonlinear Processing with Thermal Noise

- **Threshold Detection**

Probability of detection with thermal noise:

$$P_D = \frac{1}{2} \text{erfc} \left(\frac{V_{th} - V_{signal}}{\sqrt{2\sigma_{thermal}^2}} \right) \quad (35)$$

False alarm probability:

$$P_{FA} = \frac{1}{2} \text{erfc} \left(\frac{V_{th}}{\sqrt{2\sigma_{thermal}^2}} \right) \quad (36)$$

Optimal threshold (Neyman-Pearson):

$$V_{th}^{opt} = \frac{V_{signal}}{2} + \frac{\sigma_{thermal}^2}{V_{signal}} \ln \left(\frac{P_{FA}}{P_D} \right) \quad (37)$$

- **Rectification and Peak Detection**

Half-wave rectifier with thermal noise:

$$y(t) = \max(0, x(t) + n(t)) \quad (38)$$

Mean output:

$$\langle y \rangle = \frac{x}{\sqrt{2\pi\sigma_n^2}} \exp \left(-\frac{x^2}{2\sigma_n^2} \right) + \frac{x}{2} \text{erfc} \left(-\frac{x}{\sqrt{2\sigma_n^2}} \right) \quad (39)$$

- **Logarithmic Amplification**

Subthreshold MOSFET as log amplifier:

$$V_{out} = nV_T \ln \left(\frac{I_{in}}{I_0} \right) + v_{noise}(t) \quad (40)$$

Thermal noise contribution:

$$\sigma_{v,noise}^2 = \frac{n^2 k_B T}{q I_{in}} \cdot g_m R_{load} \quad (41)$$

Dynamic range:

$$DR = 20 \log_{10} \left(\frac{I_{max}/I_{min}}{\sigma_{noise}} \right) \quad (42)$$

7 5.7 Multi-Channel Processing

- **Beamforming with Thermal Arrays**

Array factor with thermal noise:

$$AF(\theta) = \sum_{n=0}^{N-1} w_n e^{j k n d \sin \theta} + \sum_{n=0}^{N-1} \eta_n(t) \quad (43)$$

Thermal noise correlation matrix:

$$\mathbf{R}_n = \sigma_{thermal}^2 \mathbf{I} \quad (44)$$

Optimal beamforming weights:

$$\mathbf{w}_{opt} = \frac{\mathbf{R}_n^{-1} \mathbf{a}(\theta_0)}{\mathbf{a}^H(\theta_0) \mathbf{R}_n^{-1} \mathbf{a}(\theta_0)} \quad (45)$$

- **Principal Component Analysis**

Covariance matrix with thermal noise:

$$\mathbf{R} = \mathbf{R}_{signal} + \sigma_{thermal}^2 \mathbf{I} \quad (46)$$

Eigenvalue decomposition:

$$\mathbf{R} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H \quad (47)$$

Thermal noise subspace identification:

$$\lambda_i = \sigma_{thermal}^2 \text{ for } i > r \quad (48)$$

where r is signal rank.

- **Independent Component Analysis**

FastICA with thermal constraints:

$$\mathbf{w}_{k+1} = E[\mathbf{x} g(\mathbf{w}_k^T \mathbf{x})] - E[g'(\mathbf{w}_k^T \mathbf{x})] \mathbf{w}_k \quad (49)$$

Thermal noise impact on convergence:

$$\text{Convergence rate} \propto \frac{1}{1 + \sigma_{thermal}^2 / \sigma_{signal}^2} \quad (50)$$

8 5.8 Hardware Implementation

• Thermal Noise Amplifiers

Low-noise amplifier design:

$$NF = 1 + \frac{R_s}{R_n} + \frac{I_n^2 R_s^2}{V_n^2} \quad (51)$$

Optimum source resistance:

$$R_{s,opt} = \sqrt{\frac{V_n^2}{I_n^2}} \quad (52)$$

Minimum noise figure:

$$NF_{min} = 1 + 2\sqrt{\frac{I_n^2 R_n}{V_n^2}} \quad (53)$$

• Correlation Processing Units

Analog multiplier using Gilbert cell:

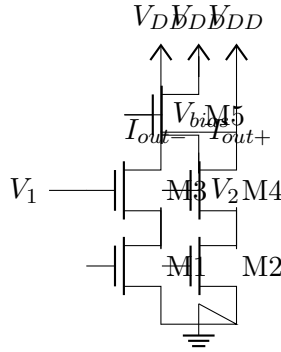


Figure 3: Gilbert cell multiplier

Output current:

$$I_{out} = I_{out+} - I_{out-} = g_m V_1 V_2 + i_{noise}(t) \quad (54)$$

• Digital Signal Processing

Fixed-point arithmetic with thermal quantization:

$$Q[x] = \text{round} \left(\frac{x + n_{thermal}}{\Delta} \right) \times \Delta \quad (55)$$

Thermal dithering benefit:

$$\sigma_{effective}^2 = \sigma_{quantization}^2 - \sigma_{thermal}^2 \quad (56)$$

for $\sigma_{thermal} < \Delta/2$.

9 Examples

Example 9.1 (Thermal Noise ADC Resolution). Calculate ENOB for thermal noise-enhanced ADC at 300K.

Given:

- Sampling capacitor: $C = 100$ fF
- Full-scale range: $V_{FS} = 1$ V
- Temperature: $T = 300$ K
- Oversampling ratio: $OSR = 64$

Solution:

Thermal noise voltage:

$$\sigma_{thermal} = \sqrt{\frac{k_B T}{C}} \quad (57)$$

$$= \sqrt{\frac{1.38 \times 10^{-23} \times 300}{100 \times 10^{-15}}} \quad (58)$$

$$= 64.5 \text{ V} \quad (59)$$

Quantization noise without thermal noise:

$$\sigma_q = \frac{\Delta}{\sqrt{12}} = \frac{V_{FS}/2^n}{\sqrt{12}} \quad (60)$$

With oversampling:

$$\sigma_{q,OS} = \frac{\sigma_q}{\sqrt{OSR}} = \frac{V_{FS}}{2^n \sqrt{12} \times 64} \quad (61)$$

For thermal noise to improve resolution:

$$\sigma_{thermal} < \sigma_{q,OS} \quad (62)$$

$$64.5 \times 10^{-6} < \frac{1}{2^n \sqrt{768}} \quad (63)$$

$$2^n > \frac{1}{64.5 \times 10^{-6} \times \sqrt{768}} = 559 \quad (64)$$

Therefore: $n > 9.13$ bits, so ENOB 9.1 bits

Example 9.2 (Stochastic Resonance Optimization). Find optimal noise level for stochastic resonance detection.

Given:

- Signal amplitude: $A = 50$ mV
- Threshold voltage: $V_{th} = 100$ mV
- Input frequency: $f = 1$ kHz

Solution:

For optimal stochastic resonance:

$$\sigma_{opt} = \frac{V_{th} - A}{\sqrt{2 \ln(2)}} \quad (65)$$

$$= \frac{100 - 50}{\sqrt{2 \ln(2)}} \text{ mV} \quad (66)$$

$$= \frac{50}{1.177} = 42.5 \text{ mV} \quad (67)$$

Signal-to-noise ratio at optimum:

$$SNR_{opt} = \frac{A^2}{2\sigma_{opt}^2} \quad (68)$$

$$= \frac{(50)^2}{2 \times (42.5)^2} = 0.69 = -1.6 \text{ dB} \quad (69)$$

Counter-intuitively, optimal detection occurs at negative SNR!

10 Homework Problems

Assignment 5.1: Thermal Noise Processing Systems

1. Stochastic Resonance Analysis:

- Design optimal noise source for 10 mV signal detection
- Calculate probability of detection vs noise level
- Implement hardware noise generator

2. Thermal ADC Design:

- Design 12-bit ADC using thermal dithering
- Calculate required oversampling ratio
- Optimize power consumption vs resolution

3. Correlation Processing:

- Implement real-time cross-correlator
- Design for 1 s correlation window
- Calculate thermal noise floor effects

4. Adaptive Filtering:

- Design LMS filter with thermal constraints
- Find convergence time vs step size
- Analyze steady-state MSE

5. Nonlinear Processing:

- Design threshold detector with thermal noise
- Optimize ROC curve performance
- Calculate detection probability

6. Multi-Channel Arrays:

- Design 8-element beamformer
- Account for thermal noise correlation
- Optimize array geometry

7. Spectral Analysis:

- Implement FFT with thermal noise floor
- Calculate dynamic range limitations
- Design windowing for optimal SNR

8. System Integration:

- Design complete thermal processing chain
- Optimize noise budgets across stages
- Calculate overall system performance

Next Chapter Preview: Chapter 6 will cover "Ultra-Low Voltage Circuit Design" including subthreshold operation and energy optimization techniques.